ETH zürich

Mobile Computing

Wi-Fi Simulator Jemula802

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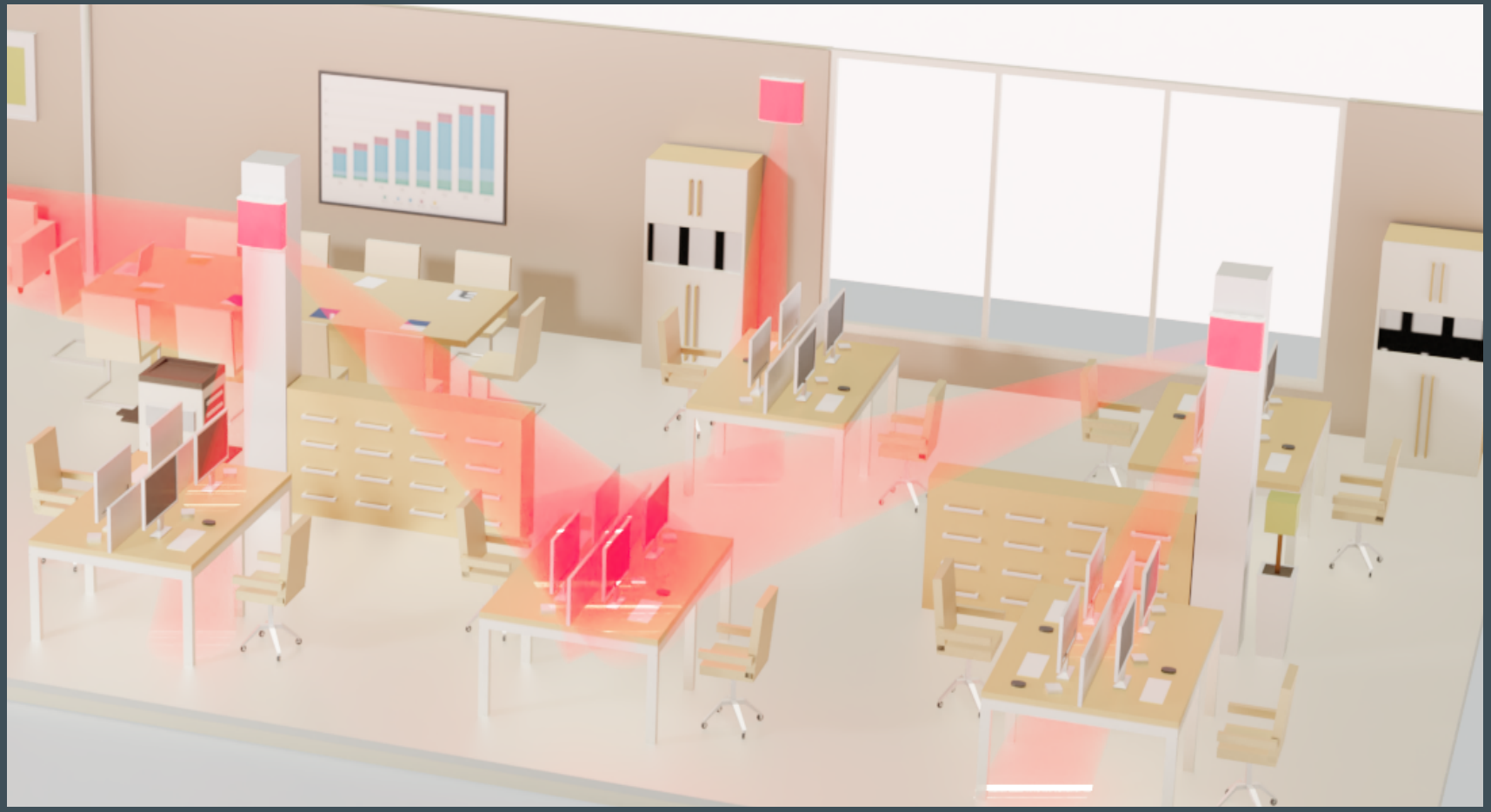
Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

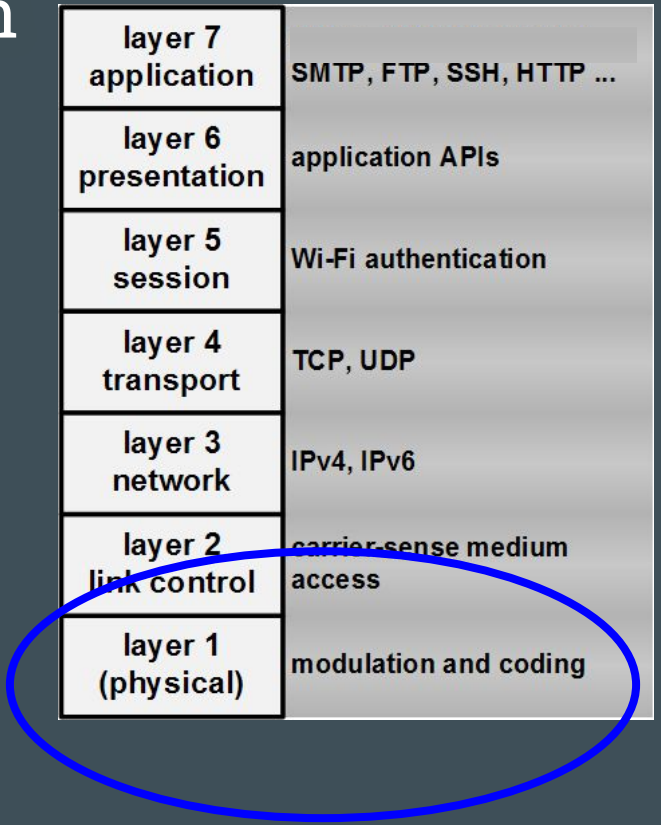
Tournament

Wi-Fi

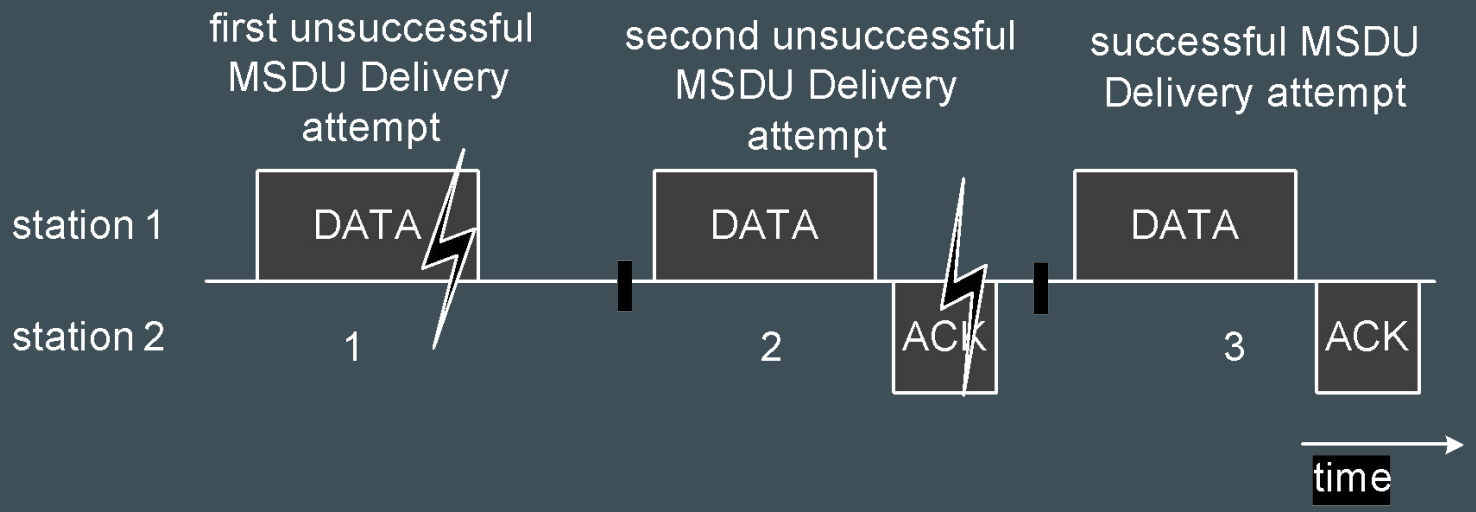


ISO/OSI Reference Model - Layer 1 and Layer 2

- ISO: International Organization for Standardization
- OSI: Open Systems Interconnection
- Wi-Fi IEEE 802.11 is for layer 1 and layer 2



802.11 Example Frame Exchange

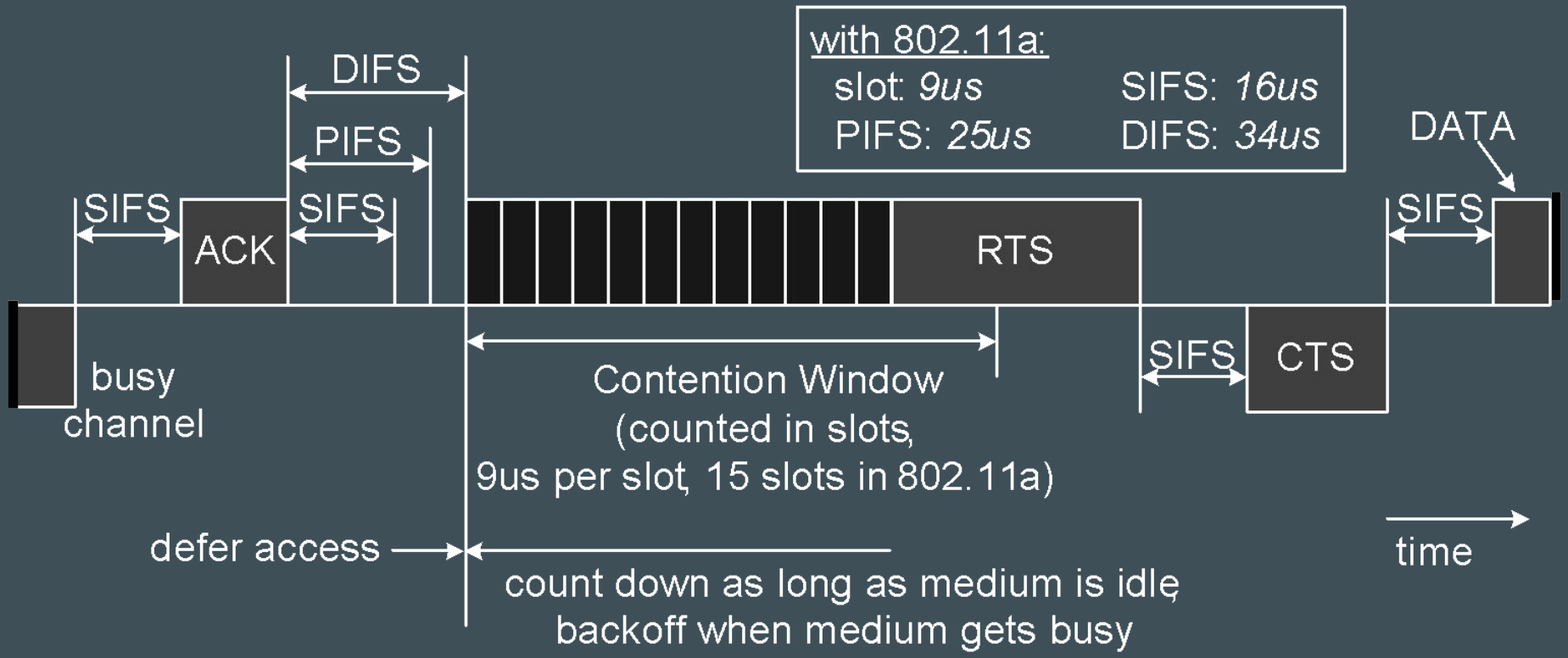


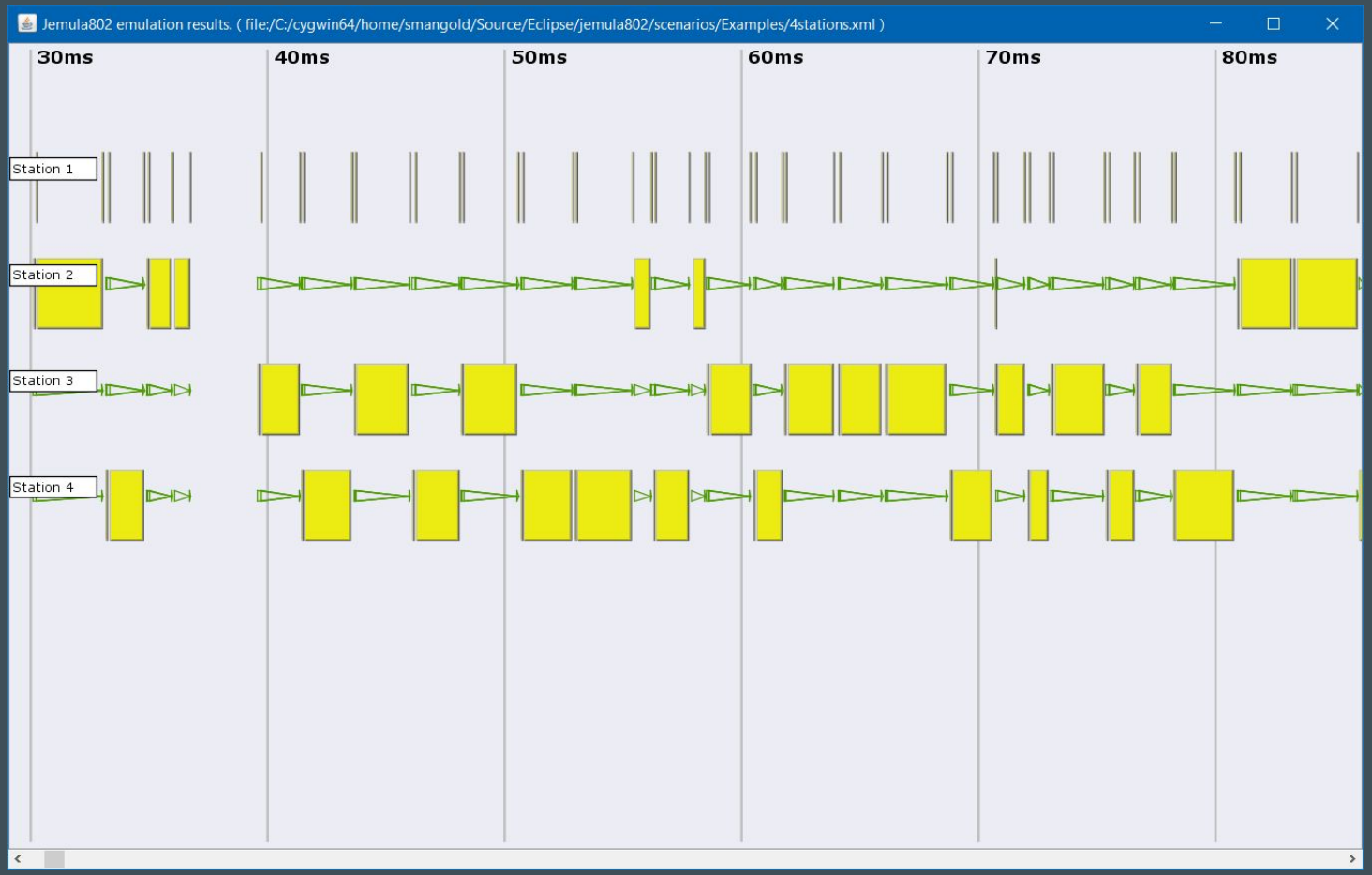
- First attempt: DATA frame not received no ACK
- Second attempt: DATA successfully received, ACK is sent but missed by station 1 (might be because of interference)
- Third and successful attempt: DATA and ACK are received

Distributed Coordination Function (DCF)

- Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)
  - Listen-before-talk principle and backoff procedure
- CCA carrier sensing is used to detect idle channel
  - CCA = Clear Channel Assessment
  - Collision avoidance (not: Collision Detection)
- Packet transmission initiated after certain idle time:
  - SIFS = Short Interframe Space (no CCA carrier sensing)
  - PIFS = PCF Interframe Space = SIFS + aSlotTime (with CCA)
  - DIFS = DCF Interframe Space = PIFS + aSlotTime (with CCA)
- SIFS is needed for transceiver turnaround

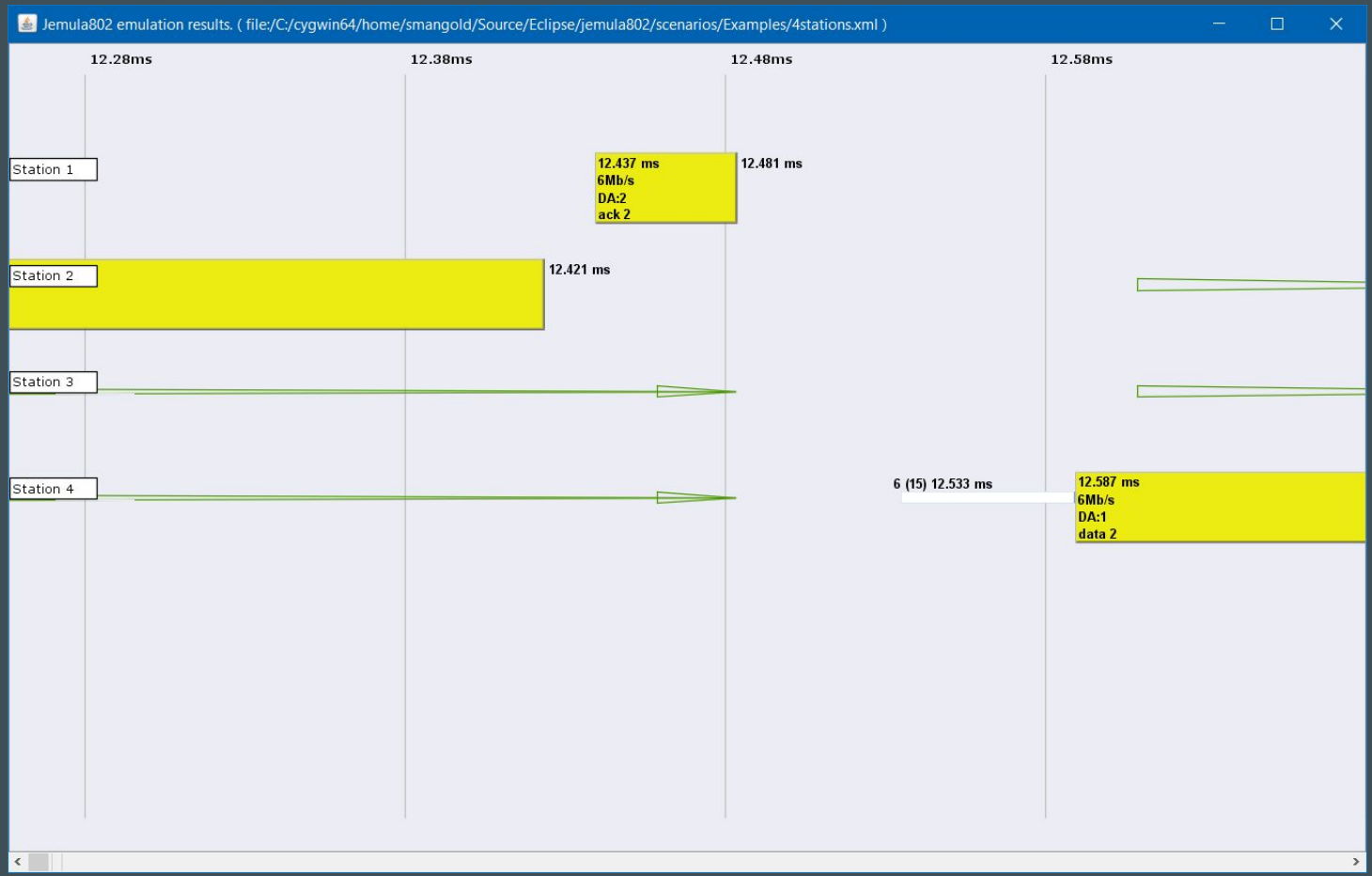
DCF Illustrated

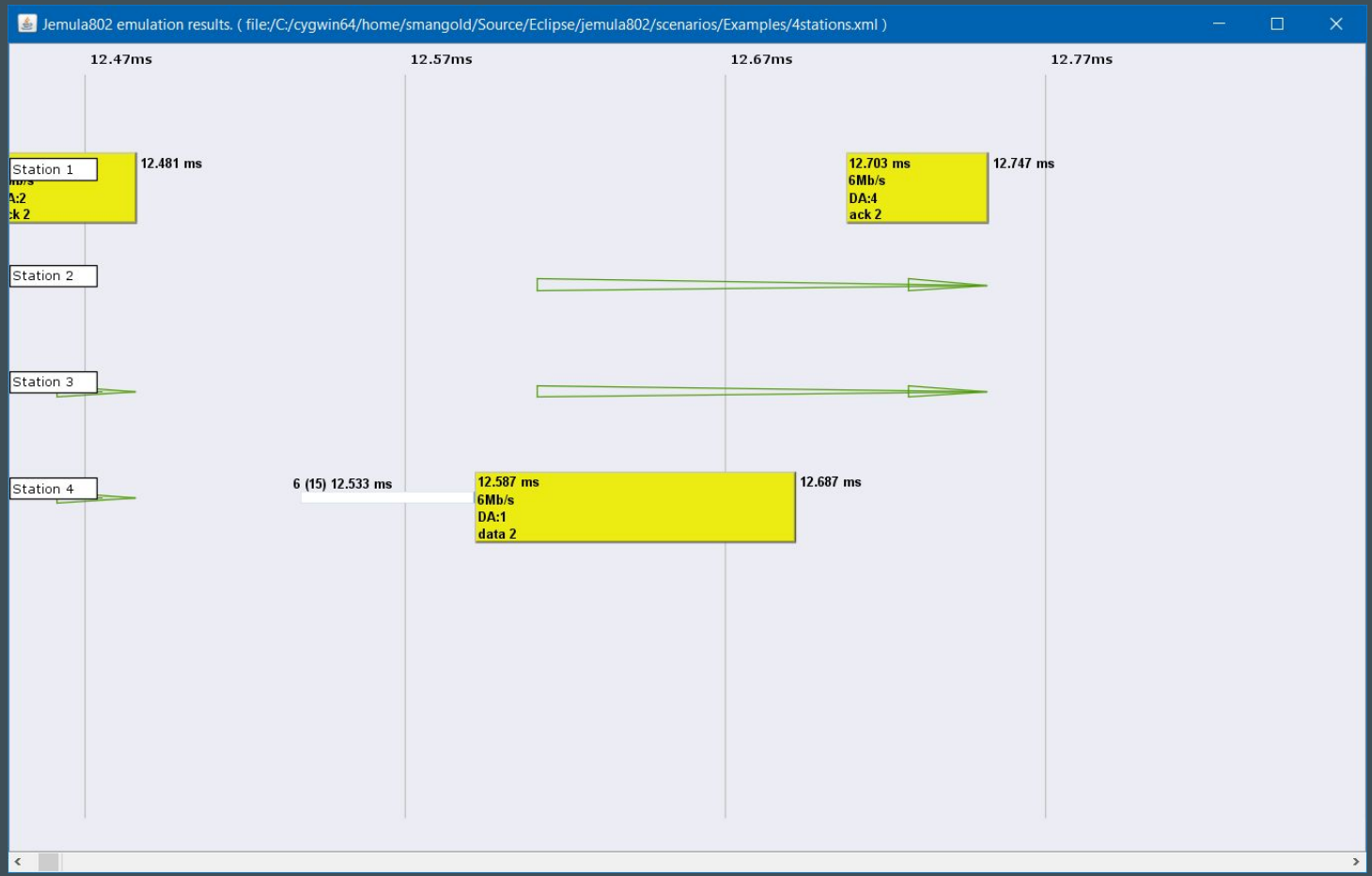


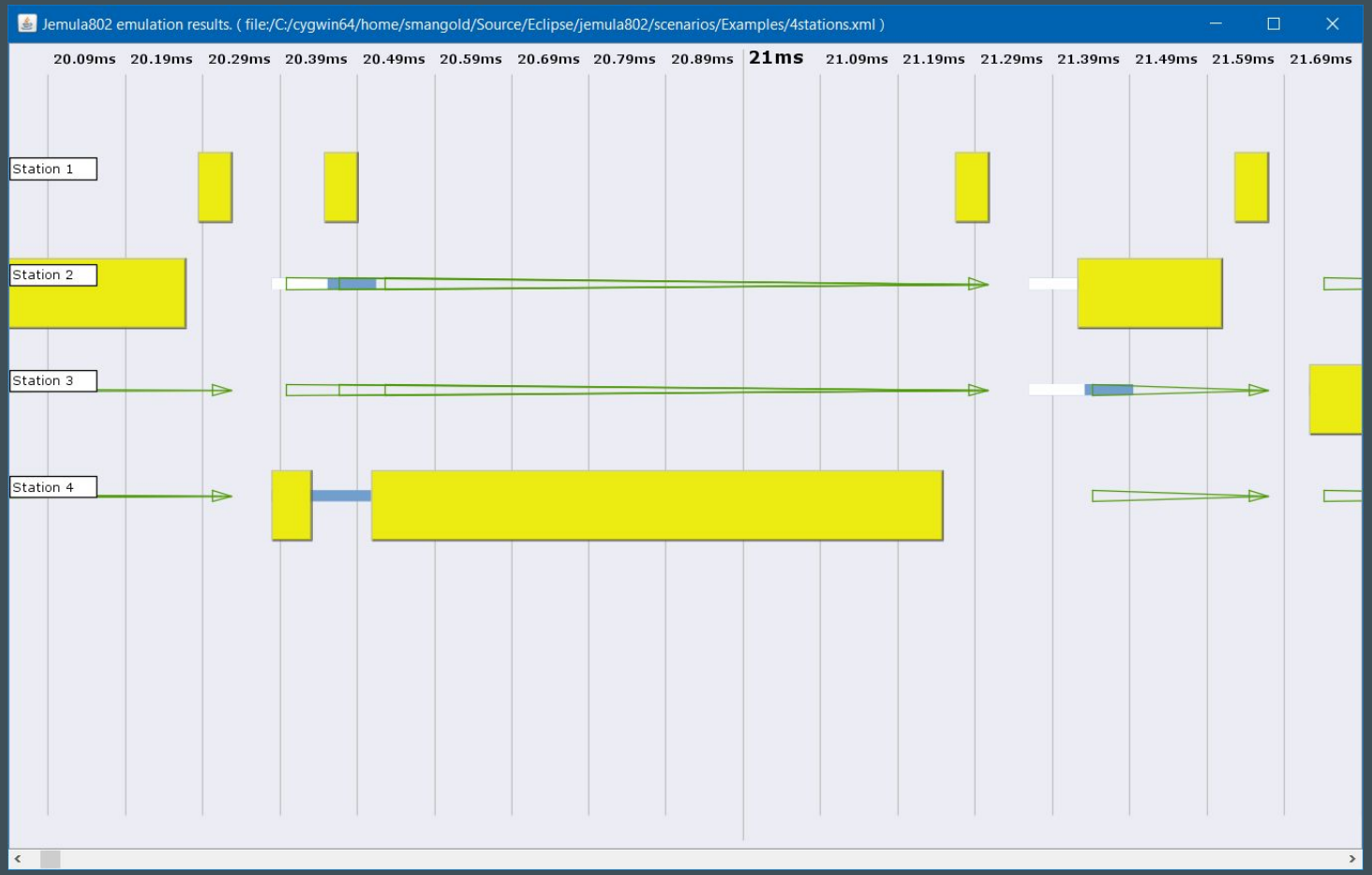

Jemula802 emulation results. (file:/C:/cygwin64/home/smangold/Source/Eclipse/jemula802/scenarios/Examples/4stations.xml)
30ms
40ms
50ms
60ms
70ms
80ms
Station 1
Station 2
Station 3
Station 4

How to use the GUI

- If the GUI is enabled in the scenario file, a Java frame will pop up and show the ongoing simulation
  - To enable, set the “showGui” attribute in the “JE802Control” tag
- Use the mouse or these keys to navigate:
  - Use the mouse wheel to scroll left and right, or the keys < and Y
  - Use left click to zoom in (right click to zoom out), or I/+ and O/-
  - F switches through different visualization of the packets
  - B toggles backoff visualization
  - N toggles NAV visualization
  - C switches through all available channels
- More details are shown if zoomed to a certain level


Jemula802 emulation results. (file:/C:/cygwin64/home/smangold/Source/Eclipse/jemula802/scenarios/Examples/4stations.xml)
12.28ms
12.38ms
12.48ms
12.58ms
Station 1
Station 2
Station 3
Station 4
12.437 ms
6Mb/s
DA:2
ack 2
12.481 ms
12.421 ms
6 (15) 12.533 ms
12.587 ms
6Mb/s
DA:1
data 2


Jemula802 emulation results. (file:/C:/cygwin64/home/smangold/Source/Eclipse/jemula802/scenarios/Examples/4stations.xml)
12.47ms
12.57ms
12.67ms
12.77ms
Station 1
12.481 ms
12.703 ms
6Mb/s
DA:4
ack 2
12.747 ms
Station 2
Station 3
Station 4
6 (15) 12.533 ms
12.587 ms
6Mb/s
DA:1
data 2
12.687 ms

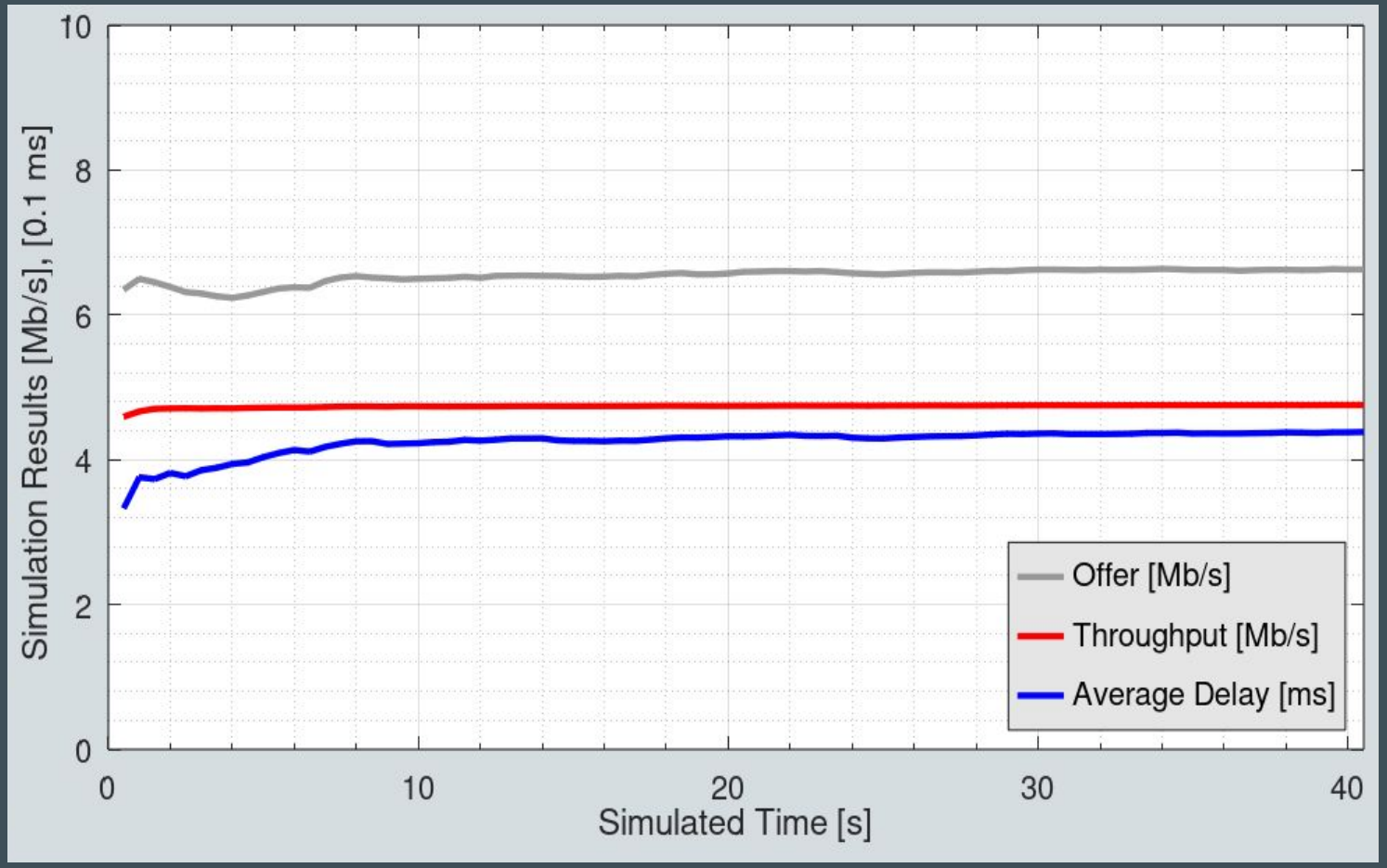

Jemula802 emulation results. (file:/C:/cygwin64/home/smangold/Source/Eclipse/jemula802/scenarios/Examples/4stations.xml)
20.09ms
20.19ms
20.29ms
20.39ms
20.49ms
20.59ms
20.69ms
20.79ms
20.89ms
21ms
21.09ms
21.19ms
21.29ms
21.39ms
21.49ms
21.59ms
21.69ms
Station 1
Station 2
Station 3
Station 4

Jemula802 - Performance Indicators

- Throughput & Delay
- Per stream, per device, or per network(s)
- Results depend on the offer (the data that needs to be communicated through the network)
- A simulation will create a folder to store all the result files
  - The folder name can be specified in the XML scenario file_ “Path2Results” attribute in the “JE802Control/JE802StatEval” tag)

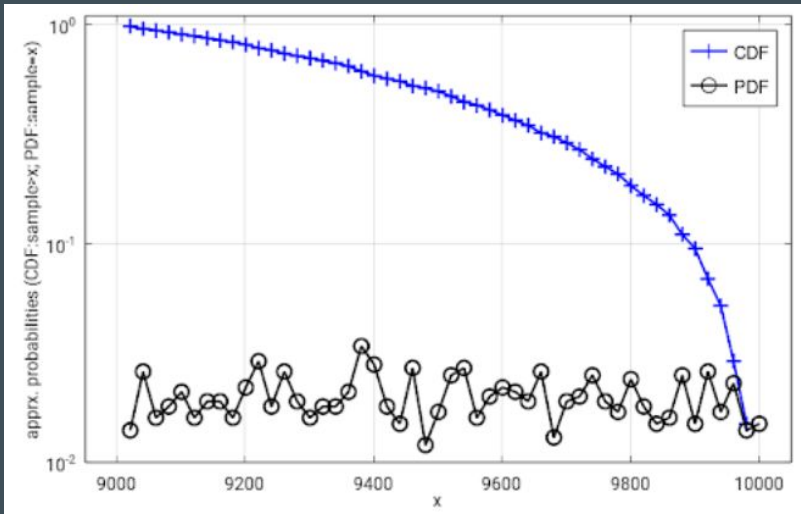
Jemula802 - Results

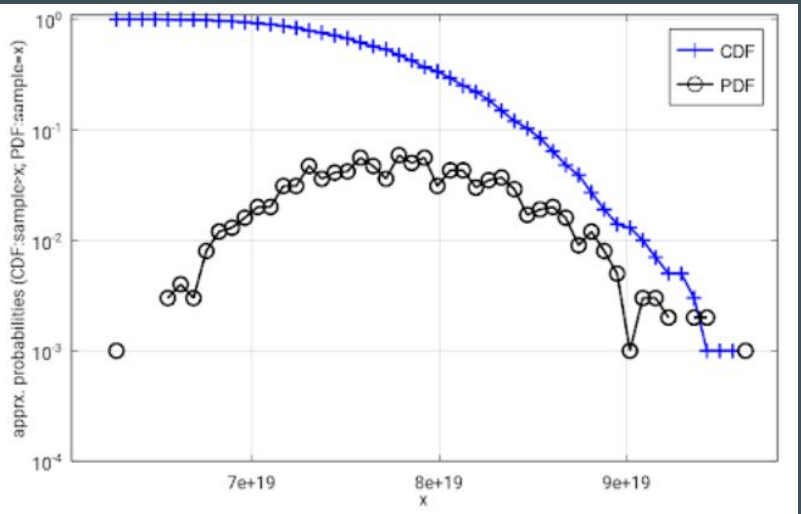
- The following result files are generated:
  - overall throughput, offer and delay (total_.m & total_.csv)
  - throughput, offer and delay per access category (_ACx.m & _ACx.csv)
  - throughput, offer and delay per link and access category
  - power consumption (overall and per station)
  - retransmission rates (only for TCP traffic)
- The *.m file are formatted as a matrix an can directly be read into Matlab or GNU Octave
- The *.csv files contain a header line with the column names


Simulation Results [Mb/s], [0.1 ms]
Simulated Time [s]
Offer [Mb/s]
Throughput [Mb/s]
Average Delay [ms]
0
2
4
6
8
10
0
10
20
30
40

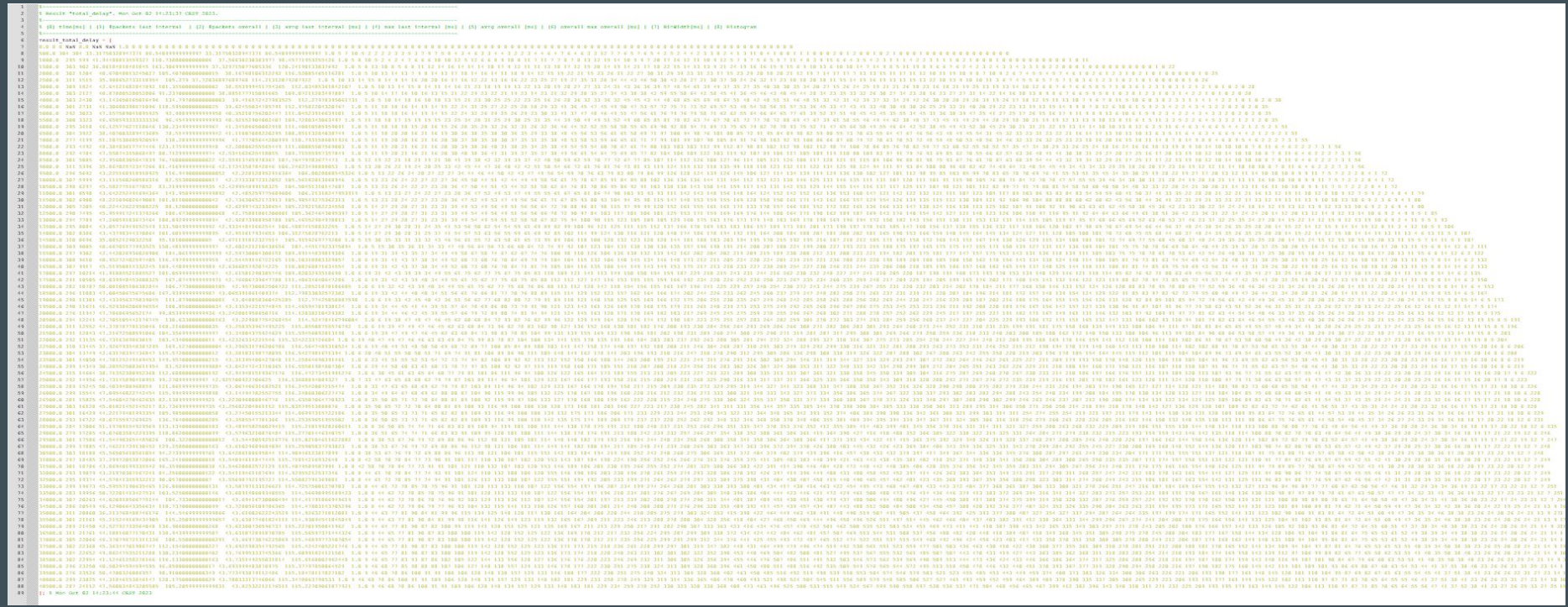
Do not only show the average of resulting delays

- Use distributions instead
- Jemula802 produces histograms in the delay results





Jemula802 Delays with Histogram

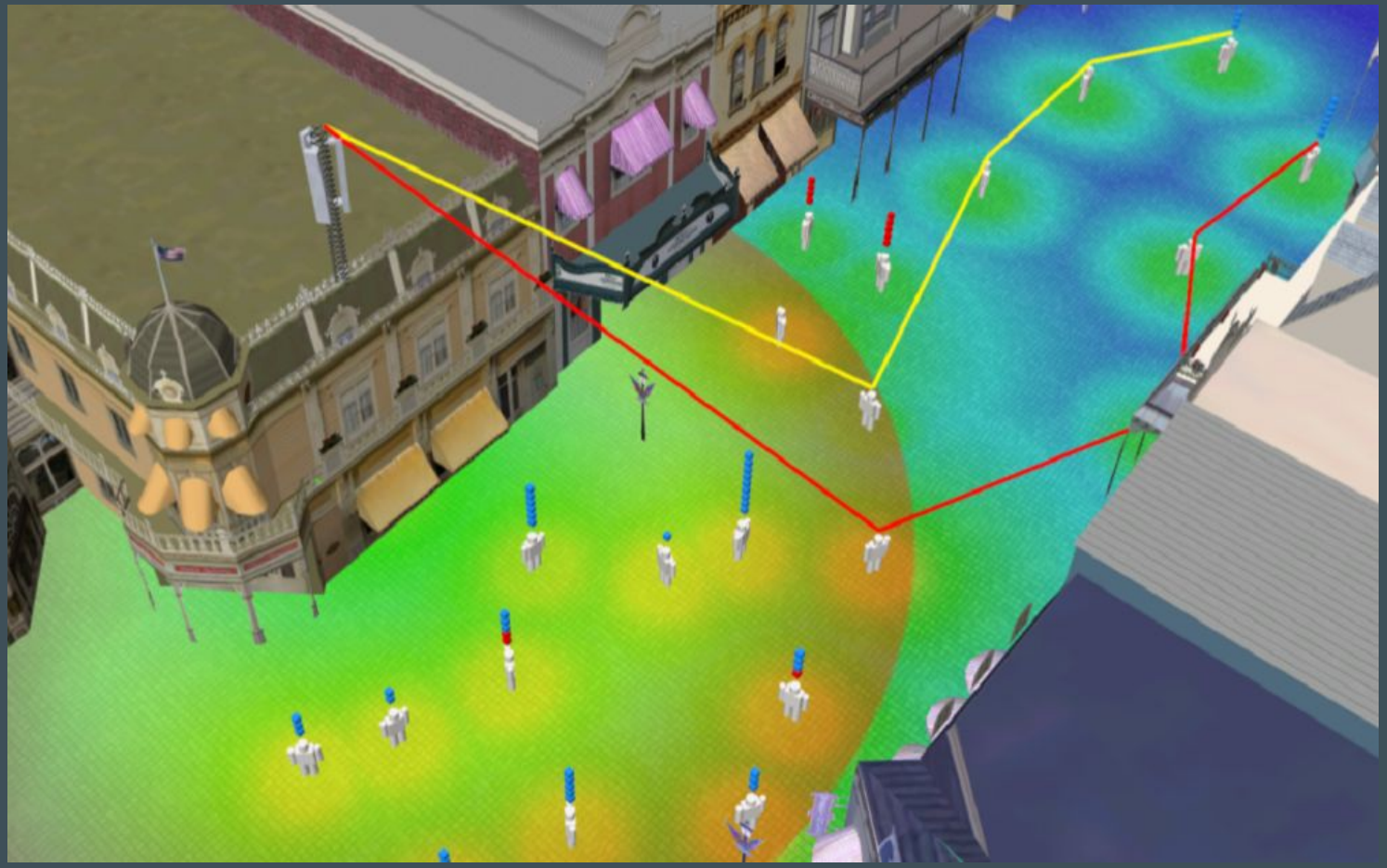


Jemula802 Delays with Histogram

```
%=============================================================================================================================
% Result "total_delay". Mon Oct 02 14:23:37 CEST 2023.
%-----------------------------------------------------------------------------------------------------------------------------
% (0) time[ms] | (1) #packets last interval  | (2) #packets overall | (3) avrg last interval [ms] | (4) max last interval [ms] | (5) avrg overall [ms] | (6) overa
%=============================================================================================================================
result_total_delay = [
0.0 0 0 NaN 0.0 NaN NaN 1.0 0 0 
```

Jemula802 - Google Earth Visualization

- Jemula802 generates a Google Earth file (*.kmz) if enabled in the XML scenario file
  - Attribute “generateGoogleEarth” in “JE802Control/JE802Animation” tag
- The file can be found in the result folder after the simulation has completed
  - The different parts of the visualization can be enabled using the check-boxes on the left
  - Hit the play button in the top left corner of the map. If it plays too fast, adjust the speed in the settings of the play bar



Correlation of Signals

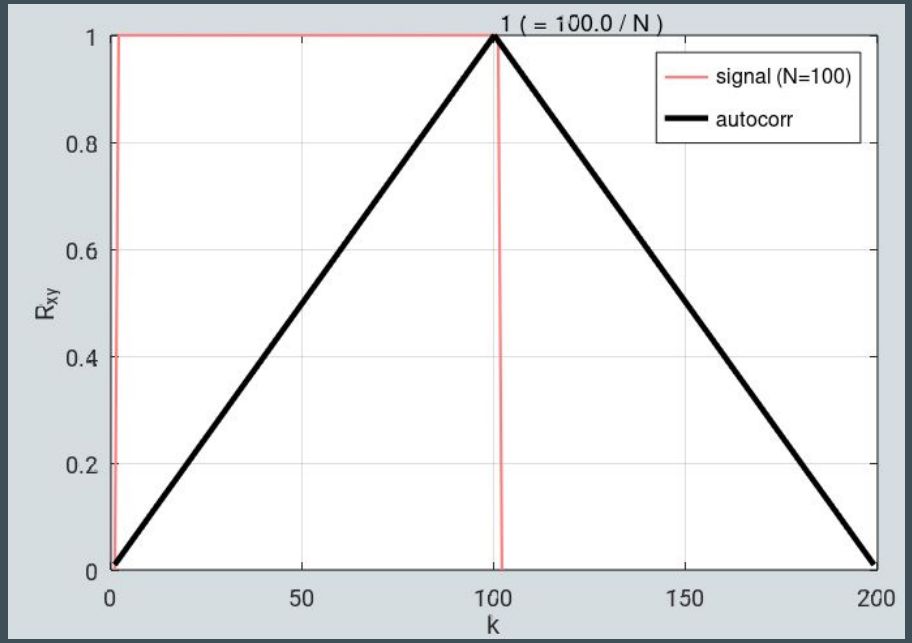
- The cross-correlation estimate between vectors "x" and "y" (of length N) for lag "k" is given by

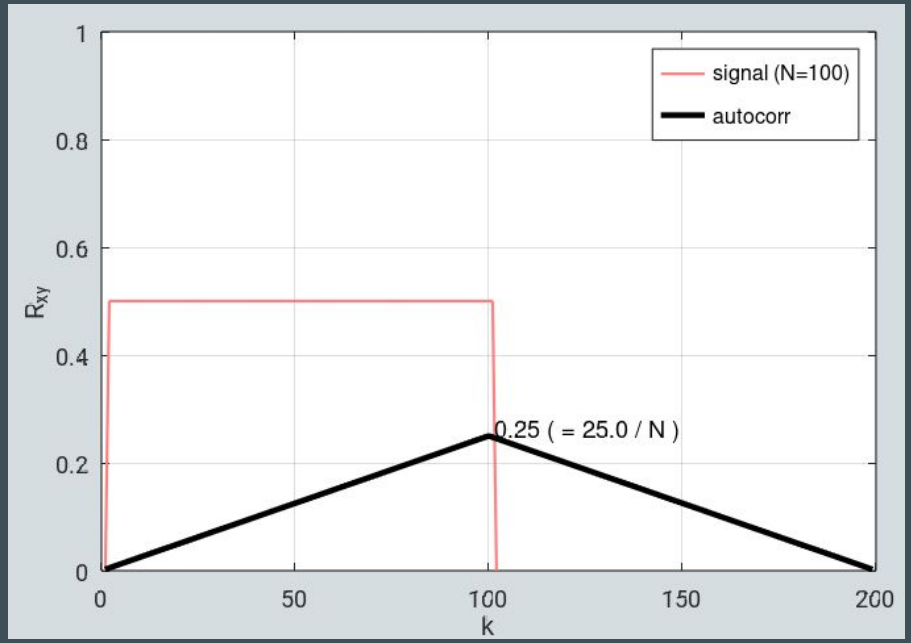
$$R_{xy}(k) = \sum_{i=1}^{N} x_{i+k} \cdot \mathrm{conj}(y_i)$$

- The term “conj” affects only the imaginary part and can be ignored for real numbers

Correlation of Signals

- Autocorrelation estimate: vectors x = y





Correlation of Signals

- With random values:

